

OpenClaw AI Pet Tutorial

Before starting, please complete the [01-Hardware Configuration], [02-openclaw API-KEY Configuration], [03-openclaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solution Configuration Tests] under the `Prerequisites` directory. If you are currently only using TUI / whatsapp, you can skip `04` for now. This article assumes the basic environment has been prepared and only expands on the AI pet project itself.

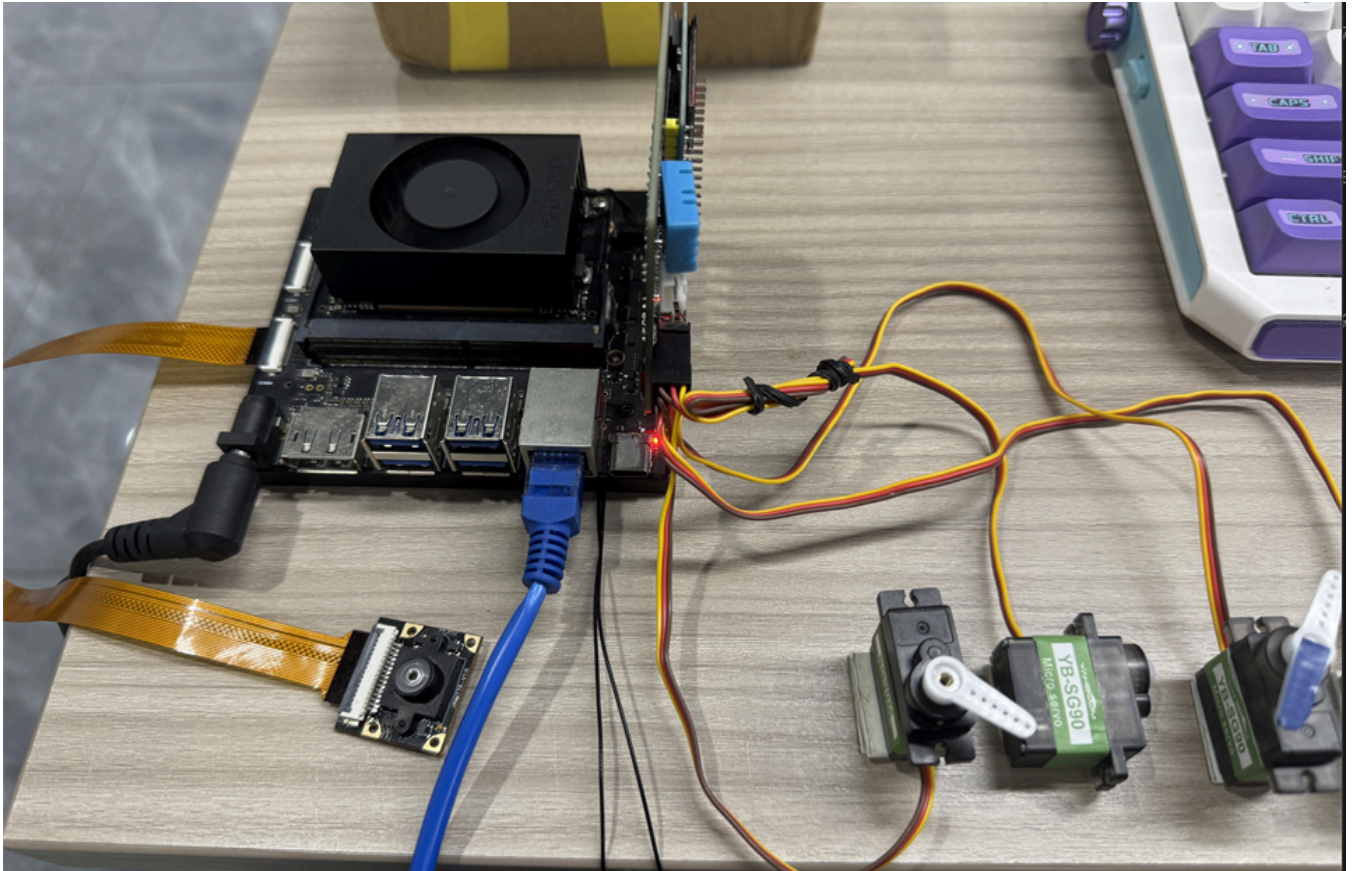
1. Tutorial Objectives

This article demonstrates OpenClaw's interaction capability as a desktop AI pet through the combination of "action + light + screen + dialogue". After completing this tutorial, readers can:

- Understand how to make an AI pet with emotional feedback using servos, RGB, and OLED
- Communicate with OpenClaw through three methods: whatsapp, TUI, and voice code
- Complete four project scenarios: greeting, emotional expression, companionship Q&A, sleep and wake-up

2. Hardware Preparation

- OpenClaw mainboard
- Servo `[Recommended at least 1]`
- RGB light strip / RGB light bead module
- OLED display
- `[Optional] AI voice module + speaker`
- Wiring diagram



3. Software Preparation

Code entry points that will be used in the tutorial

- Servo control command: `/home/jetson/hardware_hal_demo/cli/servo`
- Voice entry: `/home/jetson/voice_to_openc1aw/src/main.py`

This project does not have a fixed set of "pet scripts", but lets OpenClaw orchestrate several basic peripheral actions:

- Servo is responsible for nodding, shaking head, swinging
- RGB is responsible for expressing mood and status
- OLED is responsible for displaying name, expressions, or short text
- Voice dialogue is responsible for making the pet more like an interactive character

4. Test Project Basic Capabilities

Before starting this project, please complete one `openc1aw tui` basic dialogue self-check; if you haven't done it yet, see [05-Three Dialogue Solution Configuration Tests]. After passing, enter AI pet project specific testing.

Enter OpenClaw TUI:

```
openc1aw tui
```

```
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main

connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)
```

Then test several basic capabilities separately:

- Let the servo nod
- Change the light to pink breathing light
- Display "Hello" on OLED

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
whatsapp	Remote interaction, classroom demonstrations	Suitable for demonstrating "remote pet raising"	Demonstrate character interaction capability
TUI	Local debugging, action verification	Easiest to check if actions and feedback match	First run through character action logic
Voice code	Final interaction experience	Most like a real talking desktop pet	As final demonstration solution

If optional AI voice module + speaker are selected, whatsapp dialogue, tui, whatsapp dialogue need to broadcast OpenClaw's reply content. You can run the following program in terminal. AI voice module dialogue does not need to run repeatedly

- Modify whether to enable broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter controls whether to broadcast. After modification and saving, run the following program in terminal
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) jetson@yahboom:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 whatsapp Solution

For whatsapp connection and robot creation steps, please directly refer to [05-Three Dialogue Solution Configuration Tests]. After creation, you can remotely interact with the AI pet in whatsapp, for example:

- Take a photo to see if there is dog food. If yes, display smile expression on OLED, make servo dance, and light up ambient lights on OLED. If no dog food, display sad expression on OLED.

20:31



< 2



+86 153 3885 7526 (你)

给自己发消息

Take a photo to see if there is dog food. If yes, display smile expression on OLED, make servo dance, and light up ambient lights on OLED. If no dog food, display sad expression on OLED.

20:31 ✓✓



I can see the scene, but there's **no dog food** visible in the image. I can see a desk with various items like a pink mug, green fan, water bottle, and a curriculum schedule, but no dog food detected

20:31 ✓✓

Dog food check completed!

Result: No dog food detected in the scene





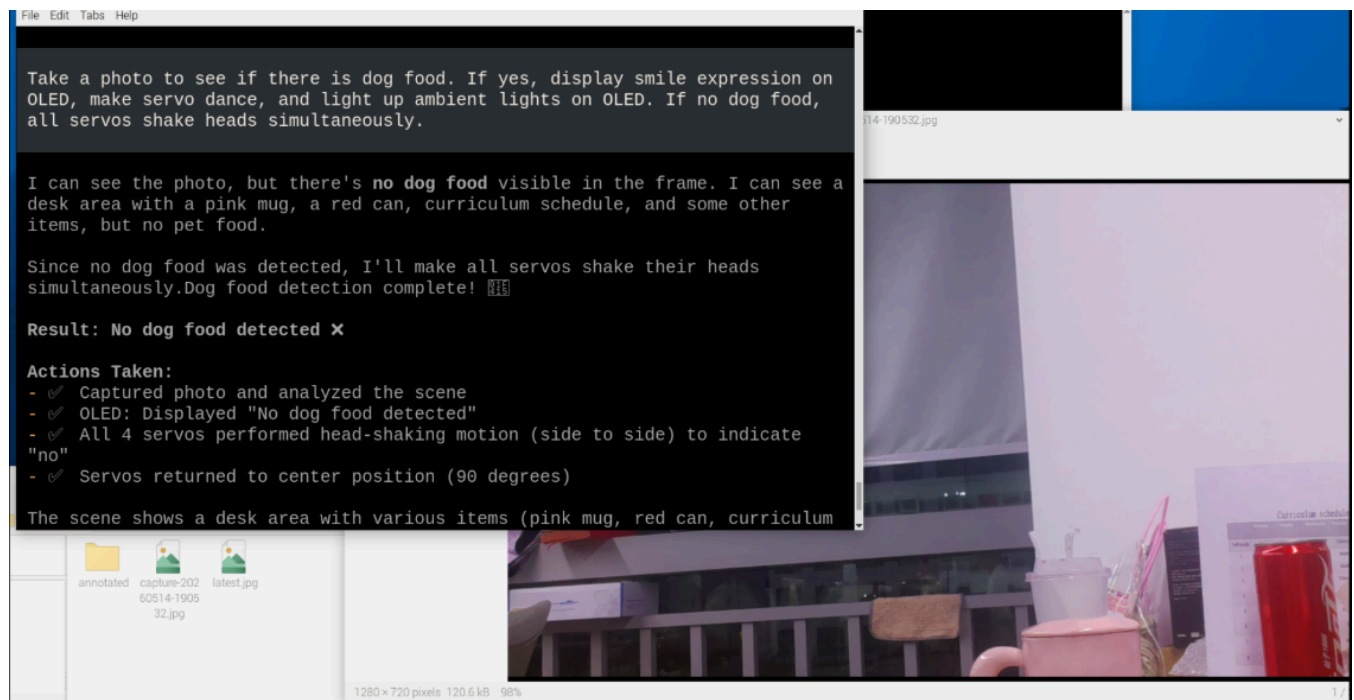
5.2 TUI Solution

If you have completed the TUI self-check in Chapter 0, directly enter terminal text dialogue:

```
openclaw tui
```

Then enter project-related commands, for example:

- Take a photo to see if there is dog food. If yes, display smile expression on OLED, make servo dance, and Light up ambient Lights on OLED. If no dog food, all servos shake heads simultaneously.



5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration]. After waking up, you can directly say control content, for example:

- Take a photo to see if there is dog food. if yes display smile expression on OLED, if no dog food display sad expression.

```
[你] Take a photo to see if there is dog food if yes display smile expression on  
old if no dog food display sad expression.  
[发送] 正在交给 OpenClaw  
[完成] OpenClaw 返回 200  
  
[OpenClaw] ## 📷 Dog Food Detection Complete!  
  
**Result:** ✖ **No dog food  
found**  
  
**What I see in the image:**  
- Window with gray roller  
blind  
- Pink mug and red Coca-Cola can  
- Curriculum schedule paper  
-  
Some boxes and containers on shelves  
- Tissue box  
  
**No pet food or  
dog food detected!** 😞  
  
**Feedback:**  
- OLED displaying sad face:  
`:(`  
- RGB lights set to dim purple (sad mood)  
  
Want me to check  
again later or look for something else? 📷  
[信息] 回复已裁剪为播报文本: 437 -> 121 字
```

6. Project Implementation

Note: The following are all dialogue demonstrations using `openclaw_tui` method. You can also choose whatsapp or voice method to complete.

6.1 Scenario One: Greeting

Experimental Objective

Let the AI pet make action and text feedback when the user sends a greeting.

Effect Description

The servo can make a simple nodding or head-turning action, OLED displays welcome text, RGB lights up soft colors. This way, the feeling of the pet "appearing" will be very obvious.

Example Instructions

- Let the little pet greet me
- Let the pet welcome me

6.2 Scenario Two: Emotional Expression

Experimental Objective

Let the AI pet express states like happy, quiet, angry, sleepy using lights and screen.

Effect Description

This scenario is most suitable for reflecting "character sense". For example, when happy, light warm breathing light; when sleepy, light blue dim light; when quiet, only display brief status.

Example Instructions

- Let the pet enter happy state
- Draw simple expressions on OLED

6.3 Scenario Three: Companionship Q&A

Experimental Objective

Let the AI pet not only move, but also respond to questions like a desktop companion character.

Effect Description

OpenClaw can first complete the dialogue, then combine servo nodding, shaking head and screen text to make "speaking" and "action" happen simultaneously, looking more like a character with personality.

Example Instructions

- First answer my question, then nod
- If you agree, express with actions
- Chat with me about today's mood

6.4 Scenario Four: Sleep and Wake-up

Experimental Objective

Let the AI pet have the ability to switch between standby, sleep, and wake-up states.

Effect Description

In sleep mode, most lights can be turned off, keeping only low-brightness state; when waking up, restore lights, screen, and action feedback, making the project more complete.

Example Instructions

- Let the pet enter sleep mode
- Wake up the pet